Waves

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Q1. The equation of a plane progressive wave is given by $y = 5\cos\pi(200t - x/150)$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s.

NTA B.Tech · 2 Apr 2026, Shift 1 (Q35)

(1) 120

(2) 150

(3) 200

(4) 300

✓ Answer: (4) 300

Solution:

Comparing with $y = A\cos(\omega t - kx)$: $\omega = 200\pi$, $k = \pi/150$.

Wave velocity $v = \omega/k = 200\pi \div (\pi/150) = 200 \times 150 = 30000$ cm/s = 300 m/s.

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in